

EL TORO MCAS, CA, USA

WMO: 722908

Lat: 33.667N

Lon: 117.733W

Elev: 384

StdP: 14.49

Time Zone: -8.00 (NAP)

Period: 90-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 43.2	(c) 45.3	(d) 10.4	(e) 9.5	(f) 59.4	(g) 15.4	(h) 12.1	(i) 59.8	(j) 23.6	(k) 59.9	(l) 19.5	(m) 59.2	(n) 2.9	(o) 90	(p) 0.327

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 19.6	(c) 91.9	(d) 67.8	(e) 88.9	(f) 67.5	(g) 86.0	(h) 66.8	(i) 71.7	(j) 86.1	(k) 70.4	(l) 84.5	(m) 69.2	(n) 82.8	(o) 6.0	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.4	98.5	79.1	65.0	93.8	77.8	63.9	90.1	76.3	35.7	86.3	34.4	85.2	33.4	82.9	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 15.5	(b) 12.3	(c) 10.6	(e) 38.8	(f) 99.3	(g) 2.4	(h) 2.6	(i) 37.1	(j) 101.2	(k) 35.7	(l) 102.7	(m) 34.3	(n) 104.2	(o) 32.6	(p) 106.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.2	57.7	58.3	59.7	63.0	65.3	68.9	72.4	74.4	73.0	68.6	63.0	57.4
	DBStd	7.46	4.55	5.11	4.53	4.81	4.12	4.11	3.23	3.90	4.11	5.18	5.43	4.21
	HDD50	7	1	3	0	0	0	0	0	0	0	0	0	3
	HDD65	1111	233	199	175	94	47	11	0	0	0	16	99	237
	CDD50	5544	239	235	301	389	476	566	695	756	690	575	389	233
	CDD65	1172	6	10	10	33	57	127	230	291	241	126	38	3
	CDH74	8313	74	112	133	356	313	792	1366	2155	1689	917	369	37
	CDH80	2652	7	20	27	80	79	237	350	776	618	358	97	3
Wind	WSAvg	3.8	3.9	4.3	4.1	4.0	4.0	3.9	3.6	3.5	3.2	3.6	3.7	4.1
Precipitation	PrecAvg	11.8	2.1	2.6	2.4	0.9	0.2	0.1	0.0	0.0	0.2	0.2	1.5	1.8
	PrecMax	25.6	7.5	13.6	10.2	3.7	1.6	0.3	0.4	0.8	1.7	1.7	6.7	7.9
	PrecMin	3.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	5.6	2.0	3.2	2.2	1.1	0.4	0.1	0.1	0.2	0.4	0.4	1.5	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.7	83.3	83.5	87.1	87.9	92.9	91.1	96.8	94.2	95.3	88.3	78.0
		MCWB	56.2	59.0	59.1	62.5	65.1	67.6	69.0	71.4	68.8	65.0	60.6	54.0
	2%	DB	75.6	77.4	77.5	81.9	81.1	86.2	86.9	91.4	90.2	88.7	82.1	73.7
		MCWB	53.4	56.6	58.2	60.2	64.1	67.2	69.7	70.0	68.7	63.8	57.9	54.3
	5%	DB	71.5	72.0	72.8	78.1	76.9	81.5	83.6	87.9	86.9	83.4	78.0	70.0
		MCWB	52.8	54.9	57.5	60.6	62.9	65.9	68.4	69.1	68.3	62.2	56.5	52.8
	10%	DB	67.7	67.5	68.9	73.8	73.9	78.0	81.2	84.4	83.4	78.5	74.1	66.3
		MCWB	51.6	54.7	56.6	59.4	61.4	64.6	67.4	68.6	68.0	62.5	55.3	52.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.4	61.9	62.7	66.3	67.7	70.5	72.7	74.7	73.1	70.1	63.8	60.2
		MCDB	65.9	75.6	75.9	80.5	82.9	86.3	87.1	89.6	86.5	87.8	79.5	69.2
	2%	WB	58.3	59.3	60.6	64.0	65.3	68.4	70.6	72.2	71.0	67.6	61.4	57.3
		MCDB	63.9	69.4	72.4	77.3	78.6	83.2	83.9	87.2	85.6	82.6	75.6	65.4
	5%	WB	57.1	57.8	59.1	62.3	63.6	66.7	69.2	70.8	69.4	65.5	59.8	55.9
		MCDB	63.5	66.4	69.4	74.3	75.2	80.2	81.4	84.5	84.2	77.8	72.5	64.6
	10%	WB	55.5	56.4	57.6	60.3	62.1	65.1	68.0	69.4	68.1	63.8	58.3	54.6
		MCDB	63.3	64.2	65.8	71.3	73.1	77.5	79.5	82.7	82.0	74.6	69.0	63.3
Mean Daily Temperature Range	5% DB	MDBR	17.3	17.2	17.5	19.0	16.2	17.6	17.6	19.6	19.4	19.2	19.7	17.9
		MCDBR	24.6	25.4	25.1	26.4	21.2	24.2	21.3	24.0	25.3	26.9	25.4	23.3
		MCWBR	13.6	13.5	13.5	13.7	11.3	12.1	10.0	11.2	12.7	14.3	14.4	13.7
	5% WB	MCDBR	17.4	18.6	19.8	21.9	18.3	21.9	19.2	20.7	22.2	21.6	20.5	17.9
		MCWBR	12.3	11.5	11.7	12.8	10.0	11.2	9.3	10.5	11.7	11.9	13.1	12.1
Clear-Sky Solar Irradiance	taub		0.313	0.321	0.345	0.354	0.353	0.367	0.376	0.362	0.346	0.344	0.305	0.303
	taud		2.555	2.545	2.457	2.436	2.473	2.438	2.479	2.505	2.539	2.520	2.620	2.622
	Ebn at Noon		287	296	296	296	295	290	286	290	291	284	289	285
	Edh at Noon		26	29	34	36	35	36	35	33	31	29	24	23
All-Sky Solar Radiation	RadAvg		1020	1298	1734	2102	2253	2383	2371	2257	1907	1447	1127	922
	RadStd		69	127	90	107	149	145	120	57	72	103	66	69

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (49 neighbors)	+1.13	N/A	-3.06	+1.30	N/A	N/A	N/A	-202	+401	+202

Nomenclature: See separate page